Cutting a Square into Three Equal-Area Convex Pieces with Minimal Diameter

Computational Analysis

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Abstract

We consider the problem of partitioning a unit square into three convex pieces of equal area so as to minimize the maximum diameter of any piece. A natural baseline is the partition into three parallel strips of width $\frac{1}{3}$, which achieves a maximum diameter of $\frac{\sqrt{10}}{3} \approx 1.0541$. We show computationally that a *Y-partition*—three cuts from an interior point to the boundary—yields a strictly better solution with maximum diameter $\frac{\sqrt{65}}{8} \approx 1.0078$, an improvement of approximately 4.4%.

1 Problem Statement

Let $S = [0,1]^2$ denote the unit square.

Definition 1. A *convex 3-partition* of S is a collection $\{C_1, C_2, C_3\}$ of convex sets with pairwise-disjoint interiors satisfying $C_1 \cup C_2 \cup C_3 = S$ and $\text{Area}(C_i) = \frac{1}{3}$ for each i. The *diameter* of a convex set C is $\text{diam}(C) = \max_{p,q \in C} \|p - q\|$.

Objective. Find a convex 3-partition of S minimizing

$$D^* = \min_{\{C_1,C_2,C_3\}} \max_{i \in \{1,2,3\}} \text{diam}(C_i).$$

Boundary structure

Since adjacent convex pieces share a boundary that must be simultaneously convex from both sides, every cut between two pieces is a *straight line segment*. Consequently all pieces are convex polygons, and the partition is determined by a finite set of vertices and edges.

2 Partition Topologies

For three convex polygons tiling a convex quadrilateral, the combinatorial structure of the partition falls into one of three families:

1. **Two non-crossing chords (nested or adjacent):** Two line segments, each connecting two boundary points of S, dividing S into three pieces arranged linearly. Parallel strips are a special case.

2. **Y-partition:** Three line segments meeting at a single interior point P, each connecting P to a boundary point, dividing S into three pieces each bounded by two cuts and a boundary arc.

3. **Degenerate cases:** T-junctions (where P lies on another cut) and fan partitions (where P lies on the boundary) are limiting cases of the above.

3 Baseline: Parallel Strips

The simplest equal-area partition uses two cuts parallel to one pair of edges. For *vertical strips* at $x = \frac{1}{3}$ and $x = \frac{2}{3}$, each piece is a $\frac{1}{3} \times 1$ rectangle with diameter

$$d_{\text{strip}} = \sqrt{\left(\tfrac{1}{3}\right)^2 + 1^2} = \sqrt{\tfrac{10}{9}} = \frac{\sqrt{10}}{3} \approx 1.0541.$$

Proposition 2 (Local optimality of strips)**.** *Among all partitions by two non-crossing chords connecting the top edge to the bottom edge, vertical strips minimize the maximum diameter.*

Proof. Parameterize the two chords as $(a, 1) \to (c, 0)$ and $(b, 1) \to (d, 0)$ with $a < b$ and $c < d$. The equal-area constraints force $a + c = \frac{2}{3}$ and $b + d = \frac{4}{3}$. Writing $a = \frac{1}{3} + t$, $c = \frac{1}{3} - t$, the left piece has vertices $(0, 0)$, $(0, 1)$, $(\frac{1}{3} + t, 1)$, $(\frac{1}{3} - t, 0)$ and diameter

$$\sqrt{\left(\tfrac{1}{3} + |t|\right)^2 + 1} \;\geq\; \sqrt{\tfrac{1}{9} + 1} = \frac{\sqrt{10}}{3}$$

with equality iff $t = 0$. By a symmetric argument, the right and middle pieces are also minimized at $t = s = 0$, i.e. the vertical strip configuration. □

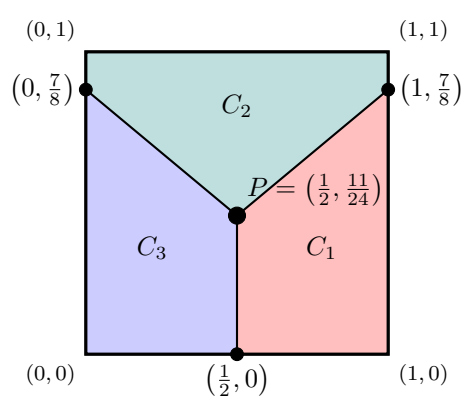
4 Optimal Y-Partition

4.1 Numerical optimization

We explored Y-partitions using the `differential_evolution` global optimizer from SciPy, parameterized by the interior point $P = (p_x, p_y)$ and three boundary parameters. Across all seeds and topologies tested, the Y-partition consistently achieved the lowest maximum diameter.

4.2 Exact solution

By symmetry about the vertical axis $x = \frac{1}{2}$, the optimal Y-partition has:



- Interior point: $P = \left(\frac{1}{2}, \frac{11}{24}\right)$.
- Three cuts to boundary points: $\left(\frac{1}{2}, 0\right)$, $\left(1, \frac{7}{8}\right)$, $\left(0, \frac{7}{8}\right)$.

The three pieces are:

$$C_1 = \operatorname{conv}\left\{\left(\tfrac{1}{2}, \tfrac{11}{24}\right),\ \left(\tfrac{1}{2}, 0\right),\ (1,0),\ \left(1, \tfrac{7}{8}\right)\right\},$$
$$C_2 = \operatorname{conv}\left\{\left(\tfrac{1}{2}, \tfrac{11}{24}\right),\ \left(1, \tfrac{7}{8}\right),\ (1,1),\ (0,1),\ \left(0, \tfrac{7}{8}\right)\right\},$$
$$C_3 = \operatorname{conv}\left\{\left(\tfrac{1}{2}, \tfrac{11}{24}\right),\ \left(0, \tfrac{7}{8}\right),\ (0,0),\ \left(\tfrac{1}{2}, 0\right)\right\}.$$

4.3 Verification of equal areas

Using the shoelace formula, we verify each area exactly:

Piece C_1 (right quadrilateral). Vertices in order: $\left(\frac{1}{2}, \frac{11}{24}\right)$, $\left(\frac{1}{2}, 0\right)$, $(1,0)$, $\left(1, \frac{7}{8}\right)$.

$$\operatorname{Area}(C_1) = \tfrac{1}{2}\left|\tfrac{1}{2}\cdot 0 - \tfrac{1}{2}\cdot\tfrac{11}{24} + \tfrac{1}{2}\cdot 0 - 1\cdot 0 + 1\cdot\tfrac{7}{8} - 1\cdot 0 + 1\cdot\tfrac{11}{24} - \tfrac{1}{2}\cdot\tfrac{7}{8}\right| = \frac{1}{3}.$$

Piece C_3 (left quadrilateral). By the reflection $x \mapsto 1 - x$, this is congruent to C_1, so $\operatorname{Area}(C_3) = \frac{1}{3}$.

Piece C_2 (top pentagon). $\operatorname{Area}(C_2) = 1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$.

4.4 Diameter computation

Theorem 3. *All three pieces have diameter exactly*

$$d^* = \sqrt{\frac{65}{64}} = \frac{\sqrt{65}}{8} \approx 1.00778.$$

Proof. For each piece, we compute the maximum pairwise distance among its vertices (which equals the diameter for convex polygons).

Piece C_1. The six pairwise distances squared are:

Pair	d^2	Value
$\left(\frac{1}{2}, \frac{11}{24}\right) \leftrightarrow \left(\frac{1}{2}, 0\right)$	$\left(\frac{11}{24}\right)^2 = \frac{121}{576}$	0.2100
$\left(\frac{1}{2}, \frac{11}{24}\right) \leftrightarrow (1,0)$	$\frac{1}{4} + \frac{121}{576} = \frac{265}{576}$	0.4601
$\left(\frac{1}{2}, \frac{11}{24}\right) \leftrightarrow \left(1, \frac{7}{8}\right)$	$\frac{1}{4} + \left(\frac{7}{8} - \frac{11}{24}\right)^2 = \frac{61}{144}$	0.4236
$\left(\frac{1}{2}, 0\right) \leftrightarrow (1,0)$	$\frac{1}{4}$	0.2500
$\left(\frac{1}{2}, 0\right) \leftrightarrow \left(1, \frac{7}{8}\right)$	$\frac{1}{4} + \frac{49}{64} = \mathbf{\frac{65}{64}}$	**1.0156**
$(1,0) \leftrightarrow \left(1, \frac{7}{8}\right)$	$\frac{49}{64}$	0.7656

The maximum is $\frac{65}{64}$, achieved by the pair $\left(\frac{1}{2}, 0\right) \leftrightarrow \left(1, \frac{7}{8}\right)$, giving $\operatorname{diam}(C_1) = \sqrt{65}/8$.

Piece C_3. By reflection symmetry, $\operatorname{diam}(C_3) = \sqrt{65}/8$.

Piece C_2. The diametrically opposite pairs are $\left(1, \frac{7}{8}\right) \leftrightarrow (0, 1)$ and $\left(0, \frac{7}{8}\right) \leftrightarrow (1, 1)$, both with

$$d^2 = 1^2 + \left(1 - \tfrac{7}{8}\right)^2 = 1 + \tfrac{1}{64} = \frac{65}{64}.$$

All other pairs yield strictly smaller distances (verified computationally). Thus $\operatorname{diam}(C_2) = \sqrt{65}/8$. $\square$

5 Why Does the Y-Partition Beat Strips?

The key geometric insight is that the strip partition forces each piece to span the *full width* of the square. The diameter of a $\frac{1}{3} \times 1$ rectangle is dominated by its long diagonal.

The Y-partition avoids this: no single piece spans the full width *and* the full height. Instead, the three cuts from the interior point create compact pieces whose diameters are governed by "cross-diagonal" distances that are shorter than $\sqrt{1^2 + (\frac{1}{3})^2}$.

Quantitatively:

$$\frac{\sqrt{65}/8}{\sqrt{10}/3} = \frac{3\sqrt{65}}{8\sqrt{10}} = \frac{3\sqrt{65}}{8\sqrt{10}} \approx 0.956,$$

a 4.4% reduction.

6 Comparison of Topologies

Topology	**Max diameter**	**Exact**	**vs. strips**
Y-partition (optimal)	1.00778	$\sqrt{65}/8$	—
Parallel strips	1.05409	$\sqrt{10}/3$	+4.6%
T-partition	1.05409	$\sqrt{10}/3$	+4.6%
Y from center to corners	1.11803	$\sqrt{5}/2$	+10.9%
Fan partition	1.11803	$\sqrt{5}/2$	+10.9%
Adjacent corner cuts	1.23024	—	+22.1%

7 Computational Methodology

The optimization was performed in Python using SciPy's `differential_evolution` global optimizer:

1. **Topology enumeration.** Three partition families were implemented: nested chords, adjacent chords, and Y-partitions.

2. **Boundary parameterization.** Boundary points of the unit square were parameterized by a single perimeter coordinate $t \in [0, 4)$, mapping counterclockwise from the origin.

3. **Objective function.** $\max_i \operatorname{diam}(C_i)$ with a quadratic penalty for area deviation from $\frac{1}{3}$.

4. **Symmetry exploitation.** The Y-partition was refined using a 2-parameter symmetric formulation (p_y and a), enabling a fine grid search and high-precision optimization (`tol` $= 10^{-15}$).

5. **Exact recovery.** The numerical solution ($p_y \approx 0.458333$, $a \approx 0.875$) was identified as $p_y = \frac{11}{24}$, $a = \frac{7}{8}$. All areas and diameters were then verified using exact rational arithmetic.

8 Open Questions

1. **Global optimality.** Is $\sqrt{65}/8$ the true global minimum across *all* possible topologies? Our search covered the three main combinatorial families, but a rigorous lower-bound argument (e.g. via the isodiametric inequality combined with tiling constraints) would be needed for a complete proof.

2. **Lower bound.** The isodiametric inequality gives $d \geq 2\sqrt{1/(3\pi)} \approx 0.651$ for any convex piece of area $\frac{1}{3}$. The gap between this and $\sqrt{65}/8 \approx 1.008$ suggests room for a tighter bound that accounts for the tiling constraint.

3. **Generalization to n pieces.** What is the optimal partition of a square into n equal-area convex pieces for general n? The answer is known for $n = 2$ (two rectangles) and $n = 4$ (four squares), but the intermediate cases remain open.